

MATH 55 SECOND MIDTERM EXAM, PROF. SRIVASTAVA
APRIL 9, 2026, 3:40PM-5:00PM, VLSB 2050.

Name: _____

SID: _____

GSI: _____

NAME OF THE STUDENT TO YOUR LEFT: _____

NAME OF THE STUDENT TO YOUR RIGHT: _____

INSTRUCTIONS: Write all answers clearly in the provided space. This exam includes some space for scratch work which will not be graded. **Do not unstaple the exam.** Write your name and SID on every page. Show your work — numerical answers without justification will be considered suspicious and will not be given full credit. Write proofs in complete English sentences. Calculators, phones, cheat sheets, textbooks, and your own scratch paper are not allowed.

UC BERKELEY HONOR CODE: *As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others.*

Sign here: _____

Question	Points
1	8
2	6
3	6
4	8
5	8
6	6
7	8
Total:	50

Do not turn over this page until your instructor tells you to do so.
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Name and SID: _____

1. Select true or false for each of the following. There is no need to provide an explanation.

(a) (2 points) For all $n \geq 1$, $\binom{2n}{n} \leq \binom{3n}{n}$.

☐ True

☐ False

(b) (2 points) If G is a simple graph with n vertices and $n - 1$ edges, then G is a tree.

☐ True

☐ False

(c) (2 points) If G is a graph with a vertex of degree 8, then G cannot be 2-colorable.

☐ True

☐ False

(d) (2 points) If G and H are isomorphic graphs, then they must have the same number of connected components.

☐ True

☐ False

[Scratch Space Below]

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2. (6 points) Prove by induction that $6 \mid n^3 - n$ for all $n \geq 1$.

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3. A tech company requires employees to generate a secure 8-character password. The password must be made using only uppercase English letters (26 options) and digits (10 options).

(a) (2 points) How many 8-character passwords can be created if there are no restrictions?

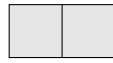
(b) (4 points) How many 8-character passwords contain at least three digits?

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4. Let $f(n)$ denote the number of ways to tile a $1 \times n$ strip using 1×1 , 1×2 , and 1×3 tiles, as illustrated below.



1×1



1×2



1×3

- (a) (4 points) Compute $f(1)$, $f(2)$, and $f(3)$, and find a recurrence relation for $f(n)$ when $n \geq 4$. Justify your recurrence.

- (b) (4 points) Prove by strong induction that $f(n) < 2^n$ for all $n \geq 1$.

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5. Consider the graph $G = (L \cup R, E)$ where $L = \{1, 2, 3, 4\}$ and $R = \{a, b, c, d\}$ and $E = \{\{x, y\} : x \in L, y \in R\}$.

(a) (1 point) Draw a picture of G .

(b) (2 points) Is G Eulerian? Justify your answer.

(c) (2 points) Is G 2-colorable? Justify your answer.

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- (d) (3 points) Find the number of simple paths of length 4 in G starting at vertex 1. Justify your answer.

6. (6 points) Prove that if G is a simple graph which contains at least two distinct spanning trees, then G must contain a cycle.

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7. (a) (5 points) Prove that if G is a graph with $n \geq 2$ vertices and exactly two vertices x and y of odd degree, then there must be a path in G connecting x and y .

- (b) (3 points) Find an example of a graph G on 4 vertices such that (i) G has exactly two vertices of even degree, x and y , and (ii) there is no path in G between x and y .

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[Scratch Paper]

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[Scratch Paper]

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[Scratch Paper]

Graph Theory Definitions

A **graph** is a pair $G = (V, E)$ where V is a finite set of **vertices** and E is a finite multiset of 2-element subsets of V , called **edges**. If E has repeated elements G is called a **multigraph**, otherwise it is called a **simple graph**. Two vertices $x, y \in V$ are **adjacent** if $\{x, y\} \in E$. An edge $e = \{x, y\}$ is said to be **incident with** x and y . The **degree** of a vertex x is the number of edges incident with it.

A **path** of length k in G is an alternating sequence of vertices and edges

$$x_0, e_1, x_1, e_2, \dots, x_{k-1}, e_k, x_k$$

where $e_i = \{x_{i-1}, x_i\} \in E$ for all $i = 1 \dots k$. The path is said to **connect** x_0, x_k , **visit** the vertices $x_0, \dots, x_k$, and **traverse** the edges $e_1, \dots, e_k$. A path is called **simple** if it contains no repeated vertices. If $x_0 = x_k$ then the path is called a **circuit**. A graph is **connected** if it contains a path between every pair of distinct vertices.

A **k-coloring** of G is a function $f : V \rightarrow \{1, \dots, k\}$ such that $f(x) \neq f(y)$ whenever x is adjacent to y . A graph is **k-colorable** if it has a k -coloring.

A graph $H = (W, F)$ is a **subgraph** of G if $W \subseteq V$ and $F \subseteq E$. If $W = V$ then H is called a **spanning** subgraph of G . A **connected component** of G is a maximal connected subgraph of G , i.e. a subgraph $H \subseteq G$ such that H is connected and adding any vertices or edges to H would make it disconnected.

A **cycle** is a circuit with no repeated edges and no repeated vertices, except the endpoints. A graph is called **acyclic** if it does not contain a cycle as a subgraph. A connected acyclic graph is called a **tree**.

An **Eulerian circuit** of a connected multigraph G is a circuit which traverses every edge of G exactly once.